

Electromagnetic Wave Simulation and Inversion Test

for Inframind Research Applicants

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Interview Questions for Research Applicant

This test evaluates the applicant's understanding of electromagnetic wave simulation, inverse modeling, and GPU-based scientific computing, with emphasis on both theoretical clarity and practical implementation.

1 Purpose

This test is designed to assess three abilities:

- understanding of electromagnetic (EM) wave simulation and inversion,
- ability to turn theory into working code,
- ability to improve efficiency using GPU acceleration.

This is not a test with a single correct answer. We care about whether you can make reasonable modeling choices, explain them clearly, and build a working solution.

2 Background

Ground-penetrating radar (GPR) and related EM methods work by sending an EM pulse into a material and recording the returned signals. If the material contains objects with different electrical properties, such as steel bars inside concrete, the wavefield will change.

In this test, you will use:

- **FDTD forward modeling:** a time-domain numerical method for simulating EM wave propagation,
- **Adjoint-state inversion:** an efficient way to compute how the model should be updated so that simulated signals better match observed signals,
- **CUDA acceleration (bonus):** moving the most computationally expensive parts of the algorithm onto a GPU.

3 Part A: Theory discussion

You should be prepared to explain the following topics clearly.

3.1 FDTD-based EM forward modeling

Please explain:

- the purpose of FDTD,
- the Yee-grid idea,
- the main time-stepping loop,
- the update equations for electric and magnetic fields,
- source injection, boundary, and numerical stability.

3.2 Adjoint-state inversion from EM data

Please explain:

- the inversion objective function,
- why directly perturbing every grid is too expensive,
- how the adjoint-state method computes gradients efficiently,
- the meaning of forward wavefield, residual, adjoint wavefield, and gradient,
- how model parameters (e.g., permittivity and conductivity) are updated.

3.3 GPU/CUDA acceleration

Please explain:

- how CUDA can be used to speed up FDTD-based forward modeling and adjoint-state-based inversion; in particular, identify the main computational operations that should be moved to the GPU and explain your reasoning,
- what practical issues arise, such as memory usage, memory access patterns, and CPU–GPU communication.

4 Part B: Coding task

4.1 Forward simulation

Build a **2D** or **2.5D** EM forward model for a reinforced concrete scanning problem, as illustrated in Fig. 1.

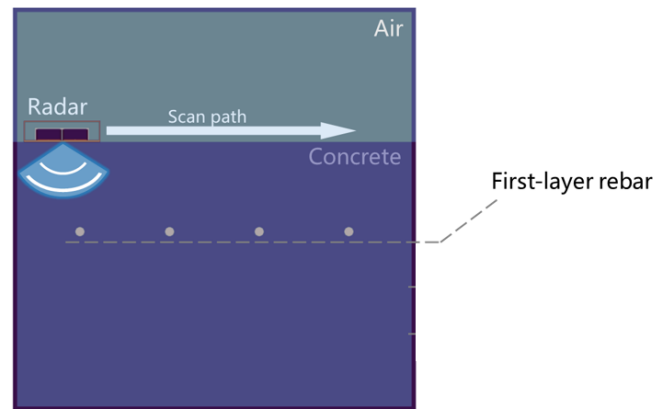


Figure 1. Example scanning setup for reinforced concrete.

Your model should include, at minimum:

- a concrete background,
- one or more rebars embedded in concrete,
- a moving transmitter–receiver scanning path,
- a time-domain source wavelet.

You must clearly define and justify your choices for:

- geometry parameters,
- material parameters,
- scanning parameters,
- grid spacing and time step,
- source waveform,
- absorbing boundary condition or PML.

Important note on “2.5D”: A simplified 2D implementation (for example TM_z or TE_z) is acceptable.

Required outputs:

- a plot of the ground-truth geometry,
- the simulated radar data (for example, a B-scan image),
- one animation of EM wave propagation.

4.2 Adjoint-state inversion

Use your simulated data to recover the geometry of the reinforced concrete model.

At minimum, your inversion should try to recover:

- the approximate location of the rebars,
- the approximate shape and size of the rebars.

You may use reasonable prior assumptions to reduce the solution space, for example:

- rebars are circular or nearly circular,
- the scene contains only two main materials (concrete and steel),
- permeability is fixed,
- conductivity or permittivity is fixed for one class and only geometry is inverted.

You may also use multiple transmitters and receivers if this improves inversion quality.

Required outputs:

- the initial model used for inversion,
- the inverted model,
- the ground-truth model,
- comparison between simulated and target signals,
- at least one quantitative error measure,
- a brief discussion of why the inversion worked well or why it failed.

4.3 Bonus: CUDA acceleration

This part is optional but strongly encouraged.

Accelerate all or part of your forward and adjoint workflow using CUDA.

Required outputs for the bonus part:

- runtime comparison between CPU and GPU versions,
- hardware information,
- a brief note on which operations you moved to the GPU.

5 Evaluation Criteria

We will mainly evaluate:

- **Completeness:** Does the code cover both forward simulation and inversion?
- **Technical understanding:** Does the applicant understand the theory behind the implementation?
- **Modeling quality:** Are the assumptions, parameters, and priors reasonable and clearly explained?
- **Inversion quality:** Does the recovered geometry match the ground truth?
- **Efficiency:** Is there evidence of profiling, optimization, or GPU acceleration?

6 Recommended References

6.1 FDTD and EM simulation

- gprMax: <https://github.com/gprMax/gprMax>

- FDTDx: <https://github.com/ymahlau/fdtdx>
- Meep: <https://github.com/NanoComp/meep>

6.2 Adjoint-state inversion and EM inversion

- Ernst, J. R., Green, A. G., Maurer, H., and Holliger, K. (2007). *Full-waveform inversion of crosshole radar data based on 2-D finite-difference time-domain solutions of Maxwell's equations*.
- Meles, G. A., Greenhalgh, S. A., and van der Kruk, J. (2010). *A new vector waveform inversion algorithm for simultaneous updating of conductivity and permittivity parameters from combination crosshole/borehole-to-surface GPR data*.
- Meles, G. A., van der Kruk, J., Greenhalgh, S. A., Ernst, J. R., Maurer, H., and Green, A. G. (2012). *GPR full-waveform sensitivity and resolution analysis using an FDTD adjoint method*.
- Kuroda, S., Takeuchi, A., and Kim, H. J. (2007). *Full-waveform inversion algorithm for interpreting crosshole radar data: A theoretical approach*.
- Soydan, O., et al. (2024). *On the utilization of the adjoint method in microwave tomography*.

Code resources for inversion

- Fast-GPR-FWI: <https://github.com/songc0a/Fast-GPR-FWI>
- GPR-FWI-Py: <https://github.com/nephilim2016/GPR-FWI-Py>